Catalan numbers I

- The *Catalan sequence* is the sequence

$$C_0, C_1, C_2, \ldots, C_n, \ldots,$$

where

$$C_n = \frac{1}{n+1}\binom{2n}{n}, \qquad (n = 0, 1, 2, \ldots)$$

is the nth *Catalan number*.



Catalan numbers II

- The Catalan number

$$C_{n-1} = \frac{1}{n}\binom{2n-2}{n-1}$$

arose as the number of ways to divide a convex polygonal region with $n+1$ sides into triangles by inserting diagonals that do not intersect in the interior.



Catalan numbers III

Theorem

The number of sequences

$$a_1, a_2, \ldots, a_{2n} \tag{8.1}$$

of $2n$ terms that can be formed by using n $+1$'s and n -1's whose partial sums satisfy

$$a_1 + a_2 + \cdots + a_k \geq 0, \qquad (k = 1, 2, \ldots, 2n) \tag{8.2}$$

equals the nth Catalan number

$$C_n = \frac{1}{n+1}\binom{2n}{n}, \qquad (n \geq 0).$$

We call a sequence (8.1) of n $+1$'s and n -1's *acceptable* if it satisfies (8.2) and *unacceptable* otherwise.



Catalan numbers IV

1. Let A_n denote the number of acceptable sequences of n +1's and n -1's, and let U_n denote the number of unacceptable ones. The total number of sequences of n +1's and n -1's is

$$\binom{2n}{n} = \frac{(2n)!}{n!n!} = A_n + U_n.$$

2. We evaluate A_n by first evaluating U_n and then subtracting from $\binom{2n}{n}$.
3. Consider an unacceptable sequence (8.1) of n +1's and n -1's. There is a *smallest* k such that the partial sum

$$a_1 + a_2 + \cdots + a_k$$

is negative. Because k is smallest, there is an equal number of +1's and -1's preceding a_k, and we have

$$a_1 + a_2 + \cdots + a_{k-1} = 0$$



Catalan numbers V

and

$$a_k = -1.$$

In particular, k is an odd integer.

4. We now reverse the signs of each of the first k terms; that is, we replace a_i by $-a_i$ for each $i = 1, 2, \ldots, k$ and leave unchanged the remaining terms. The resulting sequence

$$a_1', a_2', \ldots, a_{2n}'$$

is a sequence of $(n+1)$ $+1$'s and $(n-1)$ -1's.

5. This process is reversible: Give a sequence of $(n+1)$ $+1$'s and $(n-1)$ -1's, there is a first instance when the number of $+1$'s exceeds the number of -1's. Reversing the signs of the $+1$'s and -1's up to that point results in an unacceptable sequence of n $+1$'s and n -1's.
6. Thus, there are as many unacceptable sequences as there are sequences of $(n+1)$ $+1$'s and $(n-1)$ -1's.



Catalan numbers VI

7. The number of sequences of $(n+1)$ $+1$'s and $(n-1)$ -1's is the number

$$\frac{(2n)!}{(n+1)!(n-1)!}.$$

Hence,

$$U_n = \frac{(2n)!}{(n+1)!(n-1)!},$$

and therefore,

$$\begin{aligned} A_n &= \frac{(2n)!}{n!n!} - \frac{(2n)!}{(n+1)!(n-1)!} \\ &= \frac{(2n)!}{n!(n-1)!}\left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= \frac{1}{n+1}\frac{(2n)!}{n!n!} \\ &= \frac{1}{n+1}\binom{2n}{n}. \end{aligned}$$



Catalan numbers VII

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□

Example

There are $2n$ people in line to get into a theater. Admission is 50 cents. Of the $2n$ people, n have a 50-cent piece and n have a \$1 dollar bill. The box office at the theater rather foolishly begins with an empty cash register. In how many ways can the people line up so that whenever a person with a \$1 dollar bill buys a ticket, the box office has a 50-cent piece in order to make change? (After every one is admitted, there will be n \$1 dollar bills in the cash register.)

Solution

$C_n = \frac{1}{n+1}\binom{2n}{n}$ if the people are regarded as "indistinguishable" and $n!n!C_n = \frac{(2n)!}{n+1}$ if the people are regarded as "distinguishable".



Catalan numbers VIII

Example

A big city lawyer works n blocks north and n blocks east of her place of residence. Every day she walks $2n$ blocks to work. How many routes are possible if she never crosses (but may touch) the diagonal line from home to office?

Solution

$2C_n = \frac{2}{n+1}\binom{2n}{n}$.



Catalan numbers IX

- We have

$$\frac{C_n}{C_{n-1}} = \frac{1}{n+1}\frac{(2n)!}{n!n!}n\frac{(n-1)!(n-1)!}{(2n-2)!} = \frac{4n-2}{n+1}.$$

Therefore the Catalan numbers satisfy a homogeneous recurrence relation of order 1,

$$C_n = \frac{4n-2}{n+1}C_{n-1}, \quad (n \geq 1), \quad C_0 = 1. \tag{8.3}$$

It follows that

$$\begin{aligned} C_1 &= 1 & C_2 &= 6/3 = 2, \\ C_3 &= 2 \cdot 10/4 = 5, & C_4 &= 5 \cdot 14/5 = 14, \\ C_5 &= 14 \cdot 18/6 = 42, & C_6 &= 42 \cdot 22/7 = 132, \\ C_7 &= 132 \cdot 26/8 = 429, & C_8 &= 429 \cdot 30/9 = 1430, \\ C_9 &= 1430 \cdot 34/10 = 4,862, & C_{10} &= 4,862 \cdot 38/11 = 16,796. \end{aligned}$$



Catalan numbers X

- We now define a new sequence of numbers

$$C_1^*, C_2^*, \ldots, C_n^*, \ldots,$$

which, in order to refer to them by name, we call the *pseudo-Catalan numbers*. We let

$$C_n^* = n! C_{n-1}, \qquad (n = 1, 2, 3, \ldots).$$

The pseudo-Catalan numbers are determined by the following recurrence relation and initial condition:

$$C_n^* = (4n - 6) C_{n-1}^*, \qquad (n \geq 2), \quad C_1^* = 1. \tag{8.4}$$

The definition formula for the Catalan number and the definition of the pseudo-Catalan numbers imply the formula

$$C_n^* = (n-1)! \binom{2n-2}{n-1} = \frac{(2n-2)!}{(n-1)!}, \qquad (n \geq 1)$$



Catalan numbers XI

for the pseudo-Catalan numbers.

Example

Let $a_1, a_2, \cdots, a_n$ be n numbers. By a *multiplication scheme* for these numbers we mean a scheme for carrying out the multiplication of $a_1, a_2, \cdots, a_n$. A multiplication scheme requires a total of $n-1$ multiplications between two numbers, each of which is either one of $a_1, a_2, \cdots, a_n$ or a partial product of them. Let h_n denote the number of multiplication schemes for n numbers.

1. We have $h_1 = 1$ (this can be taken as the definition of h_1) and $h_2 = 2$, since $(a_1 \times a_2)$ and $(a_2 \times a_1)$ are two possible schemes.
2. If $n = 3$, there are 12 schemes,

$$\begin{array}{lll} (a_1 \times (a_2 \times a_3)) & (a_2 \times (a_1 \times a_3)) & (a_3 \times (a_1 \times a_2)) \\ ((a_2 \times a_3) \times a_1) & ((a_1 \times a_3) \times a_2) & ((a_1 \times a_2) \times a_3) \\ (a_1 \times (a_3 \times a_2)) & (a_2 \times (a_3 \times a_1)) & (a_3 \times (a_2 \times a_1)) \\ ((a_3 \times a_2) \times a_1) & ((a_3 \times a_1) \times a_2) & ((a_2 \times a_1) \times a_3). \end{array}$$

Thus, $h_3 = 12$.



Catalan numbers XII

3. In general, each multiplication scheme can be obtained by listing $a_1, a_2, \ldots, a_n$ in some order and then inserting $n-1$ pairs of parentheses so that each pair of parentheses designates a multiplication of two factors.
4. But in order to derive a recurrence relation for h_n, we look at it in an inductive way. Each scheme for $a_1, a_2, \ldots, a_n$ can be gotten from a scheme for $a_1, a_2, \ldots, a_{n-1}$ in exactly one of the following ways,
   1. Take a multiplication scheme for $a_1, a_2, \ldots, a_{n-1}$ (which has $n-2$ multiplications and $n-2$ sets of parentheses) and insert a_n on either side of either factor in one of the $n-2$ multiplications. Thus, each scheme for $n-1$ numbers gives $2 \cdot 2 \cdot (n-2) = 4(n-2)$ schemes for n numbers in this way.
   2. Take a multiplication scheme for $a_1, a_2, \ldots, a_{n-1}$ and multiply it on the left or right by a_n. Thus, each scheme for $n-1$ numbers gives two schemes for n numbers in this way.



Catalan numbers XIII

e.g.

$$((a_1 \times a_2) \times ((a_3 \times a_4) \times a_5))$$
$$((a_1 \times a_2) \times ((a_6 \times (a_3 \times a_4)) \times a_5))$$
$$((a_1 \times a_2) \times (((a_3 \times a_4) \times a_6) \times a_5))$$
$$((a_1 \times a_2) \times ((a_3 \times a_4) \times (a_6 \times a_5)))$$
$$((a_1 \times a_2) \times ((a_3 \times a_4)) \times (a_5 \times a_6)))$$
$$\vdots$$

and

$$(a_6 \times ((a_1 \times a_2) \times ((a_3 \times a_4) \times a_5))),$$
$$(((a_1 \times a_2) \times ((a_3 \times a_4) \times a_5)) \times a_6).$$



Catalan numbers XIV

5. Let $n \geq 2$. We see that each of the h_{n-1} multipliccation schemes for $n-1$ numbers gives

$$4(n-2)+2 = 4n-6$$

schemes for n numbers. We thus obtain the recurrence relation

$$h_n = (4n-6)h_{n-1}, \qquad (n \geq 2),$$

which together with the initial value $h_1 = 1$, determines the entire sequence $h_1, h_2, \ldots, h_n, \ldots$. This is the same type of recurrence relation with the same initial value satisfied by the pseudo-Catalan numbers (8.4).

6. Hence,

$$h_n = C_n^* = (n-1)!\binom{2n-2}{n-1}, \quad (n \geq 1).$$



Catalan numbers XV

Example

In the preceding example, suppose that we count only those multiplication schemes in which the n numbers are listed in the order $a_1, a_2, \cdots, a_n$. Thus, for instance, $((a_2 \times a_1) \times a_3)$ is no longer counted. Let g_n denote the number of multiplication schemes with this additional restriction, Then, $h_n = n!g_n$, and hence,

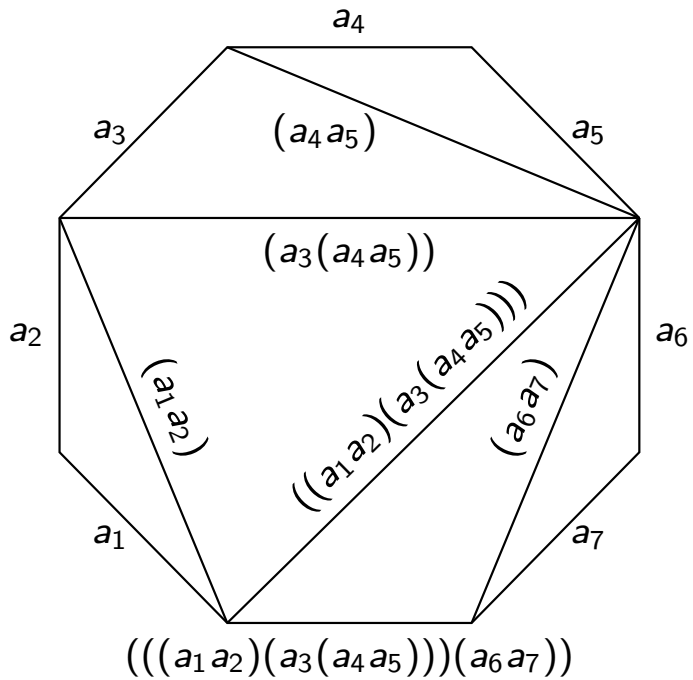
$$g_n = \frac{C_n^*}{n!} = \frac{1}{n}\binom{2n-2}{n-1} = C_{n-1}, \quad (n \geq 1), \tag{8.5}$$

showing that g_n is the $(n-1)$st Catalan number.

- The number of ways to divide a convex polygonal region with $n+1$ sides into triangle regions by inserting diagonals that do not intersect in the interior is the same as the number of multiplication schemes for n numbers given in a specified order with the common value equal to the common value equal to the $(n-1)$st Catalan number.



Catalan numbers XVI



The correspondence between the multiplication schemes for the $n = 7$ numbers and triangularizations of a convex polygonal regions of $n + 1 = 8$ sides.



Difference sequence and Stirling numbers I

- Let

$$h_0, h_1, h_2, \ldots, h_n, \ldots \tag{8.6}$$

be a sequence of numbers. We define a new sequence

$$\Delta h_0, \Delta h_1, \Delta h_2, \ldots, \Delta h_n, \ldots, \tag{8.7}$$

called the (*first order*) *difference sequence* of (8.6), by

$$\Delta h_n = h_{n+1} - h_n, \qquad (n \geq 0).$$

The terms of the difference sequence (8.7) are the differences of consecutive terms of the sequence (8.6). We can form the difference sequence of (8.7) and obtain the *second-order difference sequence*

$$\Delta^2 h_0, \Delta^2 h_1, \Delta^2 h_2, \ldots, \Delta^2 h_n, \ldots.$$



Difference sequence and Stirling numbers II

Hence,

$$\begin{aligned}\Delta^2 h_n &= \Delta(\Delta h_n)\\ &= \Delta h_{n+1} - \Delta h_n\\ &= (h_{n+2} - h_{n+1}) - (h_{n+1} - h_n)\\ &= h_{n+2} - 2h_{n+1} + h_n, \qquad (n \geq 0).\end{aligned}$$

More generally, we can inductively define the pth-*order difference sequence* of (8.6) by

$$\Delta^p h_0, \Delta^p h_1, \Delta^p h_2, \ldots, \Delta^p h_n, \ldots \quad (p \geq 1),$$

where

$$\Delta^p h_n = \Delta(\Delta^{p-1} h_n).$$

We define the 0th-*order difference sequence* of a sequence to be itself; that is

$$\Delta^0 h_n = h_n, \qquad (n \geq 0).$$



Difference sequence and Stirling numbers III

- The difference table for the sequence (8.6) is obtained by listing the pth-order difference sequences in a row for each $p = 0, 1, 2, \ldots$

$$
\begin{array}{ccccccccccc}
h_0 & & h_1 & & h_2 & & h_3 & & h_4 & & \cdots \\
& \Delta h_0 & & \Delta h_1 & & \Delta h_2 & & \Delta h_3 & & \cdots & \\
& & \Delta^2 h_0 & & \Delta^2 h_1 & & \Delta^2 h_2 & & \cdots & & \\
& & & \Delta^3 h_0 & & \Delta^3 h_1 & & \cdots & & & \\
& & & & \cdots & & & & & &
\end{array}
$$



Difference sequence and Stirling numbers IV

Example

Let a sequence $h_0, h_1, h_2, \cdots, h_n, \cdots$ be defined by

$$h_n = 2n^2 + 3n + 1, \qquad (n \geq 0).$$

The difference table for this sequence is

$$\begin{array}{ccccccccccccc}
1 & & 6 & & 15 & & 28 & & 45 & & 66 & & 91 & & \cdots \\
& 5 & & 9 & & 13 & & 17 & & 21 & & 25 & & \cdots & \\
& & 4 & & 4 & & 4 & & 4 & & 4 & & \cdots & & \\
& & & 0 & & 0 & & 0 & & 0 & & \cdots & & & \\
& & & & & & \cdots & & & & & & & &
\end{array}$$

Difference sequence and Stirling numbers V

Theorem

Let the general term of a sequence be a polynomial of degree p in n

$$h_n = a_p n^p + a_{p-1} n^{p-1} + \cdots + a_1 n + a_0, \qquad (n \geq 0).$$

Then $\Delta^{p+1} h_n = 0$ for all $n \geq 0$.

- Suppose that g_n and f_n are the general terms of two sequences, and, indeed, if c and d are constants, it follows that

$$\Delta^p(cg_n + df_n) = c\Delta^p g_n + d\Delta^p f_n, \qquad (n \geq 0) \tag{8.8}$$

for each integer $p \geq 0$. We refer to the property in (8.8) as the *linearity property* of differences.

Example

Let $g_n = n^2 + n + 1$ and let $f_n = n^2 - n - 2$ $(n \geq 0)$.

Difference sequence and Stirling numbers VI

The difference table for the g_n's is

$$\begin{array}{ccccccccccc} 1 & & 3 & & 7 & & 13 & & 21 & & \cdots \\ & 2 & & 4 & & 6 & & 8 & & \cdots & \\ & & 2 & & 2 & & 2 & & \cdots & & \\ & & & 0 & & 0 & & \cdots & & & \end{array}$$

The difference table for the f_n's is

$$\begin{array}{ccccccccccc} -2 & & -2 & & 0 & & 4 & & 10 & & \cdots \\ & 0 & & 2 & & 4 & & 6 & & \cdots & \\ & & 2 & & 2 & & 2 & & \cdots & & \\ & & & 0 & & 0 & & \cdots & & & \end{array}$$

Let

$$h_n = 2g_n + 3f_n = 2(n^2 + n + 1) + 3(n^2 - n - 2) = 5n^2 - n - 4.$$



Difference sequence and Stirling numbers VII

Then the difference table for the h_n's is obtained by multiplying the entries for the first difference table by 2 and the entries for the second difference table by 3, and then adding corresponding entries. The result is

$$\begin{array}{ccccccccc} -4 & & 0 & & 14 & & 38 & & 72 & & \cdots \\ & 4 & & 14 & & 24 & & 34 & & \cdots & \\ & & 10 & & 10 & & 10 & & \cdots & & \\ & & & 0 & & 0 & & \cdots & & & \end{array}$$

- By its very definition, the difference table for a sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ is determined by the entries in row number 0. We next observe that the difference table is also determined by the entries along the left edge, the 0th diagonal–that is, by the numbers

$$h_0 = \Delta^0 h_0, \Delta^1 h_0, \Delta^2 h_0, \Delta^3 h_0, \ldots$$

along the leftmost diagonal of the difference table.



Difference sequence and Stirling numbers VIII

- Generally,

$$h_n = \binom{n}{p}$$

is a polynomial in n of degree p whose difference table has its 0th diagonal equal to

$$\overbrace{0, 0, \ldots, 0}^{p}, 1, 0, 0, \ldots.$$



Difference sequence and Stirling numbers IX

Theorem

The general term of the sequence whose difference table has its 0th diagonal equal to

$$c_0, c_1, c_2, \ldots, c_p, 0, 0, 0, \ldots \qquad \text{where } c_p \neq 0$$

is a polynomail in n of degree p satisfying

$$h_n = c_0\binom{n}{0} + c_1\binom{n}{1} + c_2\binom{n}{2} + \cdots + c_p\binom{n}{p}. \tag{8.9}$$

Example

Consider the sequence with general term

$$h_n = n^3 + 3n^2 - 2n + 1, \qquad (n \geq 0).$$



Difference sequence and Stirling numbers X

Computer differences we obtain

$$\begin{array}{ccccccc} 1 & & 3 & & 17 & & 49 \\ & 2 & & 14 & & 32 & \\ & & 12 & & 18 & & \\ & & & 6 & & & \end{array}$$

Since h_n is a polynomial in n of degree 3, the 0th diagonal of the difference table is

$$1, 2, 12, 6, 0, 0, \ldots.$$

Hence

$$h_n = 1\binom{n}{0} + 2\binom{n}{1} + 12\binom{n}{2} + 6\binom{n}{3}. \tag{8.10}$$



Difference sequence and Stirling numbers XI

Suppose that we want to find the partial sum

$$\sum_{k=0}^{n} h_k = 1\sum_{k=0}^{n}\binom{k}{0} + 2\sum_{k=0}^{n}\binom{k}{1} + 12\sum_{k=0}^{n}\binom{k}{2} + 6\sum_{k=0}^{n}\binom{k}{3}.$$

By formula (5.12)

$$\sum_{k=0}^{n}\binom{k}{p} = \binom{n+1}{p+1}, \tag{8.11}$$

We have

$$\sum_{k=0}^{n} h_k = 1\binom{n+1}{1} + 2\binom{n+1}{2} + 12\binom{n+1}{3} + 6\binom{n+1}{4}.$$



Difference sequence and Stirling numbers XII

Theorem

Assume that the sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ has a difference table whose 0th diagonal equals

$$c_0, c_1, c_2, \ldots, c_p, 0, 0, \ldots.$$

Then

$$\sum_{k=0}^{n} h_k = c_0 \binom{n+1}{1} + c_1 \binom{n+1}{2} + \cdots + c_p \binom{n+1}{p+1}.$$

Example

Find the sum of the fourth powers of the first n positive integers.



Difference sequence and Stirling numbers XIII

Let $h_n = n^4$. Computing differences we obtain

$$\begin{array}{ccccccccc}
0 & & 1 & & 16 & & 81 & & 256 \\
& 1 & & 15 & & 65 & & 175 & \\
& & 14 & & 50 & & 110 & & \\
& & & 36 & & 60 & & & \\
& & & & 24 & & & &
\end{array}$$

Because h_n is a polynomial in n with degree 4, the 0th diagonal of the difference table equals

$$0, 1, 14, 36, 24, 0, 0, \ldots.$$

Hence,

$$1 + 2^4 + \cdots + n^4 = \sum_{k=0}^{n} k^4$$



Difference sequence and Stirling numbers XIV

$$= 0\binom{n+1}{1} + 1\binom{n+1}{2} + 14\binom{n+1}{3} + 36\binom{n+1}{4} + 24\binom{n+1}{5}$$

- Let $h_n = n^p$. The 0th diagonal of the difference table for h_n has the form

  $$c(p,0), c(p,1), c(p,2), \ldots, c(p,p), 0, 0, \ldots,$$

  and it follows that

  $$n^p = c(p,0)\binom{n}{0} + c(p,1)\binom{n}{1} + \cdots + c(p,p)\binom{n}{p}. \tag{8.12}$$

  If $p = 0$, then $h_n = 1$, a constant, and (8.12) reduces to

  $$n^0 = 1\binom{n}{0} = 1,$$



Difference sequence and Stirling numbers XV

in particular, $c(0,0) = 1$. We also have

$$c(p,0) = 0, \quad (p \geq 1).$$

- We rewrite (8.12) by introducing a new expression. Let

$$[n]_k = \begin{cases} n(n-1)\cdots(n-k+1) & \text{if } k \geq 1 \\ 1 & \text{if } k = 0. \end{cases}$$

We also note that $[n]_{k+1} = (n-k)[n]_k$. Hence, (8.12) can be rewritten as

$$\begin{aligned} n^p &= c(p,0)\frac{[n]_0}{0!} + c(p,1)\frac{[n]_1}{1!} + \cdots + c(p,p)\frac{[n]_p}{p!} \\ &= \sum_{k=0}^{p} \frac{c(p,k)}{k!}[n]_k. \end{aligned}$$



Difference sequence and Stirling numbers XVI

Now we introduce the numbers

$$S(p,k)=\frac{c(p,k)}{k!},\qquad (0\le k\le p)$$

and in terms of them, (8.12) becomes

$$\begin{aligned} n^p &= S(p,0)[n]_0+S(p,1)[n]_1+\cdots+S(p,p)[n]_p \\ &= \sum_{k=0}^{p} S(p,k)[n]_k. \end{aligned} \tag{8.13}$$

The number $S(p,k)$ just introduced are called the *Stirling numbers of the second kind*. We have

$$S(p,0)=\begin{cases} 1 & \text{if } p=0 \\ 0 & \text{if } p\ge 1. \end{cases} \tag{8.14}$$



Difference sequence and Stirling numbers XVII

and

$$S(p,p) = \frac{c(p,p)}{p!} = 1. \quad (p \geq 0). \tag{8.15}$$

Theorem

If $1 \leq k \leq p-1$, then

$$S(p,k) = kS(p-1,k) + S(p-1,k-1).$$



Difference sequence and Stirling numbers XVIII

- We observe that were it not for the factor k in front of $S(p-1,k)$ we would have the Pascal's recurrence. We have

$$n^p = \sum_{k=0}^{p} S(p,k)[n]_k. \tag{8.16}$$

and

$$n^{p-1} = \sum_{k=0}^{p-1} S(p-1,k)[n]_k.$$

Thus

$$\begin{aligned} n^p = n \cdot n^{p-1} &= n \sum_{k=0}^{p-1} S(p-1,k)[n]_k \\ &= \sum_{k=0}^{p-1} S(p-1,k)(n-k+k)[n]_k \end{aligned}$$



Difference sequence and Stirling numbers XIX

$$
\begin{aligned}
&= \sum_{k=0}^{p-1} S(p-1,k)(n-k)[n]_k + \sum_{k=0}^{p-1} S(p-1,k)k[n]_k \\
&= \sum_{k=1}^{p} S(p-1,k-1)[n]_k + k\sum_{k=1}^{p-1} S(p-1,k)[n]_k \\
&= S(p-1,p-1)[n]_p \\
&+ \sum_{k=1}^{p-1} \big\{ S(p-1,k-1) + kS(p-1,k) \big\}[n]_k .
\end{aligned}
$$

For each k with $1 \leq k \leq p-1$, comparing the coefficient of $[n]_k$ in this expression for n^p with the coefficient of $[n]_k$ in the expression (8.16), we obtain

$$S(p,k) = kS(p-1,k) + S(p-1,k-1).$$



Difference sequence and Stirling numbers XX

- The recurrence relation given in the theorem and the initial values

$$S(p,0) = 0 \quad (p \geq 1) \qquad \text{and} \qquad S(p,p) = 1 \quad (p \geq 0)$$

from (8.14) and (8.15) determine the sequence of Stirling numbers of the second kind $S(p,k)$.

$p \backslash k$	0	1	2	3	4	5	6	7	$\cdots$
0	1								
1	0	1							
2	0	1	1						
3	0	1	3	1					
4	0	1	7	6	1				
5	0	1	15	25	10	1			
6	0	1	31	90	65	15	1		
7	0	1	63	301	350	140	21	1	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

The Pascal-like triangle of these Stirling numbers.



Difference sequence and Stirling numbers XXI

Theorem

The Stirling number of the second kind $S(p, k)$ counts the number of partitions of a set of p elements into k indistinguishable boxes in which no box is empty.

- To say the boxes are indistinguishable means that we can't tell one box from another. They all look the same. Thus the only thing that matters is what the contents of the various boxes are, not *which* box holds what.
- Let $S^*(p, k)$ denote the number of partitions of a set of p elements into k indistinguishable boxes in which no box is empty. We easily see that

$$S^*(p, p) = 1, \qquad (p \geq 0)$$

and

$$S^*(p, 0) = 0, \qquad (p \geq 1).$$



Difference sequence and Stirling numbers XXII

- Consider the set of the first p positive integers $1, 2, \ldots, p$ as the set to be partitioned. The partitions of $\{1, 2, \ldots, p\}$ into nonempty, indistinguishable boxes are of two types,
  - (i) Those in which p is in all alone in a box, and
  - (ii) those in which p is not in a box by itself. Thus the box containing p contains at least one more element.
- In the case of type (i), if we remove p from the box which contains it, we are left with a partition of $\{1, 2, \ldots, p-1\}$ into $k-1$ nonempty, indistinguishable boxes. Hence there are $S^*(p-1, k-1)$ partitions of $\{1, 2, \ldots, p\}$ of type (i).



Difference sequence and Stirling numbers XXIII

- Now consider a partition of type (ii). Suppose we remove p from the box that contains it. Since p was not all alone in its box, we are left with a partition $A_1, A_2, \ldots, A_k$ of $\{1, 2, \ldots, p-1\}$ into k nonempty, indistinguishable boxes. The partition $A_1, A_2, \ldots, A_k$ of $\{1, 2, \ldots, p-1\}$ that results upon the removal of p arises from k different partitions of $\{1, 2, \ldots, p\}$, namely, from

$$
\begin{aligned}
&A_1 \cup \{p\}, A_2, \ldots, A_k,\\
&A_1, A_2 \cup \{p\}, \ldots, A_k,\\
&\qquad \vdots\\
&A_1, A_2, \ldots, A_k \cup \{p\}.
\end{aligned}
$$

Put another way, after deleting p we can't tell which box it came from, it could have been any one of the k boxes since all boxes remain nonempty upon removal of p. It follows that there are $kS^*(p-1, k)$ partitions of $\{1, 2, \ldots, p\}$ of type (ii). Hence

$$S^*(p, k) = kS^*(p-1, k) + S^*(p-1, k-1) \quad (1 \le k \le p-1),$$



Difference sequence and Stirling numbers XXIV

which satisfies the same recurrence relation as the Stirling numbers of the second kind. □

- We determine the number $S^{\#}(p,k)$ of partitions of $\{1,2,\ldots,k\}$ into k nonempty, *distinguishable* boxes. Think of one box as colored red, one colored blue, one green, and so on. Thus,

$$S^{\#}(p,k) = k!S(p,k), \quad \text{or} \quad S(p,k) = \frac{1}{k!}S^{\#}(p,k). \tag{8.17}$$



Difference sequence and Stirling numbers XXV

Theorem

For each integer k with $0 \leq k \leq p$, we have

$$S^{\#}(p,k) = \sum_{t=0}^{k} (-1)^t \binom{k}{t} (k-t)^p,$$

and hence,

$$S(p,k) = \frac{1}{k!} \sum_{t=0}^{k} (-1)^t \binom{k}{t} (k-t)^p.$$

- Let U be the set of *all* partitions of $\{1, 2, \ldots, p\}$ into k distinguishable boxes $\mathcal{B}_1, \mathcal{B}_2, \ldots, \mathcal{B}_k$. We define k properties $P_1, P_2, \ldots, P_k$ where P_i is the property that the ith box $\mathcal{B}_i$ is empty. Let A_i denote the subset of U consisting of those partitions for which $\mathcal{B}_i$ is empty. Then

$$S^{\#}(p,k) = \left| \overline{A_1} \cap \overline{A_2} \cap \cdots \cap \overline{A_k} \right|.$$



Difference sequence and Stirling numbers XXVI

- We have

$$|U| = k^p.$$

- Let t be an integer with $1 \leq t \leq k$.

$$|A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_t}| = (k-t)^p$$

counts the number of partitions of $\{1, 2, \ldots, p\}$ into $k - t$ distinguishable boxes. Thus by the inclusion-exclusion principle we have

$$S^{\#}(p,k) = \sum_{t=0}^{k} (-1)^t \binom{k}{t} (k-t)^p.$$





Difference sequence and Stirling numbers XXVII

- The *Bell number* B_p is the number of partitions of a set of p elements into nonempty, indistinguishable boxes. The Bell numbers are just the sum of the entries in a row of the triangle of Stirling numbers of the second kind; that is

$$B_p = S(p,0) + S(p,1) + \cdots + S(p,p).$$

Theorem

If $p \geq 1$, then

$$B_p = B_0 \binom{p-1}{0} + B_1 \binom{p-1}{1} + \cdots + B_{p-1} \binom{p-1}{p-1}.$$

- We partition the set $\{1, 2, \ldots, p\}$ into nonempty, indistinguishable boxes.
- The box containing p also contains a subset X (possibly empty) of $\{1, 2, \ldots, p-1\}$. The set X has t elements where t is some integer between 0 and $p-1$.



Difference sequence and Stirling numbers XXVIII

- We can choose a set X of size t in $\binom{p-1}{t}$ ways and partition the $p-1-t$ elements of $\{1,2,\ldots,p-1\}$ which don't belong to X into nonempty, indistinguishable boxes in B_{p-1-t} ways. Hence

$$\begin{aligned} B_p &= \sum_{t=0}^{p-1} B_{p-1-t} \binom{p-1}{t} \\ &= \sum_{t=0}^{p-1} B_t \binom{p-1}{p-1-t} \\ &= \sum_{t=0}^{p-1} B_t \binom{p-1}{t}. \end{aligned}$$

□

Difference sequence and Stirling numbers XXIX

- The Stirling numbers of the second kind show us how to write n^p in terms of $[n]_0, [n]_1, \ldots, [n]_p$. The Stirling numbers of the first kind play the inverse role.

$$[n]_p = \sum_{k=0}^{p} (-1)^{p-k} s(p,k) n^k. \tag{8.18}$$

The *Stirling numbers of the first kind* are the coefficients

$$s(p,k), \qquad (0 \le k \le p)$$

that occur in (8.18). It follow that $s(p,0) = 0$ $(p \ge 1)$ and $s(p,p) = 1$ $(p \ge 0)$.

Theorem

If $1 \le k \le p-1$, then

$$s(p,k) = (p-1)s(p-1,k) + s(p-1,k-1).$$

Difference sequence and Stirling numbers XXX

- We have

$$[n]_p = \sum_{k=0}^{p} (-1)^{p-k} s(p,k) n^k, \tag{8.19}$$

and

$$[n]_{p-1} = \sum_{k=0}^{p-1} (-1)^{p-1-k} s(p-1,k) n^k.$$

- We observe that

$$\begin{aligned}
[n]_p =& [n]_{p-1}(n-(p-1)) \\
=& \sum_{k=0}^{p-1} (-1)^{p-1-k} s(p-1,k) n^{k+1} \\
&+ \sum_{k=0}^{p-1} (-1)^{p-k} (p-1) s(p-1,k) n^k \\
=& \sum_{k=1}^{p} (-1)^{p-k} s(p-1,k-1) n^k \\
&+ \sum_{k=0}^{p-1} (-1)^{p-k} (p-1) s(p-1,k) n^k.
\end{aligned}$$

Difference sequence and Stirling numbers XXXI

- Comparing the coefficient of n^k in this expression with the coefficient of n^k in the expression (8.19) we obtain

$$s(p,k) = (p-1)s(p-1,k) + s(p-1,k-1).$$

for each integer k with $1 \leq k \leq p-1$. □

Theorem

The Stirling number $s(p,k)$ of the first kind counts the number of arrangements of p objects into k nonempty circular permutations.



Difference sequence and Stirling numbers XXXII

- We refer to the circular permutations in the statement of the theorem as circle. Let $s^{\#}(p,k)$ denote the number of ways to arrange p people in k nonempty circles. We have

$$s^{\#}(p,p)=1, \qquad (p \geq 0)$$

and

$$s^{\#}(p,0)=0, \qquad (p \geq 1).$$

Thus the numbers $s^{\#}(p,k)$ satisfy the same initial conditions as the Stirling number of the first kind.

- Let people be labeled $1,2,\ldots,p$. The arrangements of $1,2,\ldots,p$ into k circles are of two types.
- Those of the first type have person p in a circle by himself; there are $S^{\#}(p-1,k-1)$ of these.



Difference sequence and Stirling numbers XXXIII

- In the second type, p is in a circle with at least one other person. These can be obtained from the arrangements of $1, 2, \ldots, p-1$ into k circles by putting person p on the left of any one of $1, 2, \ldots, p-1$. Thus each arrangement of $1, 2, \ldots, p-1$ gives $p-1$ arrangements of $1, 2, \ldots, p$ in this way, and hence there is a total of $(p-1)s^{\#}(p-1, k)$ arrangements of the second type.
- Hence

$$s^{\#}(p, k) = (p-1)s^{\#}(p-1, k) + s^{\#}(p-1, k-1),$$

which satisfies the same recurrence relation as the Stirling numbers of the first kind. It now follows that $s(p, k) = s^{\#}(p, k)$. □



Partition numbers I

- A *partition of a positive integer n* is a representation of n as an unordered sum of one or more positive integers, called *parts*. The partition of 1, 2, 3, 4, and 5 are, respectively,

$$
\begin{array}{llllll}
1; & & & & & \\
2; & 1+1; & & & & \\
3; & 2+1, & 1+1+1; & & & \\
4; & 3+1, & 2+2, & 2+1+1, & 1+1+1+1; & \\
5; & 4+1, & 3+2; & 3+1+1; & 2+2+1; & \\
 & & & 2+1+1+1; & 1+1+1+1+1. &
\end{array}
$$



Partition numbers II

- A partition of n is sometimes written as

$$\lambda = n^{(a_n)} \cdots 2^{(a_2)} 1^{(a_1)}, \tag{8.20}$$

where a_i is a nonnegative integer equal to the number of parts equal to i. When written in the form (8.20), the term $i^{(a_i)}$ is usually omitted if $a_i = 0$. In this notation, the partitions of 5 are

$$5^{(1)},\ 4^{(1)}1^{(1)},\ 3^{(1)}2^{(1)},\ 3^{(1)}1^{(2)},\ 2^{(2)}1^{(1)},2^{(1)}1^{(3)},\ 1^{(5)}.$$

- Let p_n denote the number of different partitions of the positive integer n, and for convenience, let $p_0 = 1$. The *partition sequence* is the sequence of numbers

$$p_0, p_1, \ldots, p_n, \ldots.$$

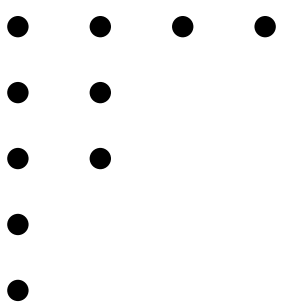
It is a simple observation that p_n equals the number of solutions in nonnegative integers $a_n, \ldots, a_2, a_1$ of the equation

$$na_n + \cdots + 2a_2 + 1a_1 = n.$$



Partition numbers III

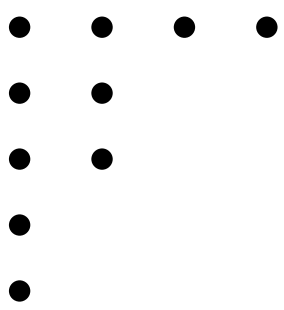
- Let λ be the partition $n = n_1 + n_2 + \cdots + n_k$ of n, where $n_1 \geq n_2 \geq \cdots \geq n_k > 0$. The *Ferrers diagram*, or simply *diagram*, of λ is a left-justified array of dots that has k rows with n_i dots in row i. For example, the diagram of the partition $10 = 4 + 2 + 2 + 1 + 1$ is



Partition numbers IV

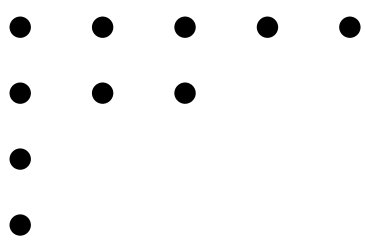
- The *conjugate partition* of the partition λ of n is the partition λ^* whose diagram is obtained from the diagram of λ by interchanging rows with columns (flipping the diagram over the diagonal running from the upper left to the lower right). For example,

A partition



$10 = 4 + 2 + 2 + 1 + 1$

Its conjugate



$10 = 5 + 3 + 1 + 1$

- If $\lambda^* = \lambda$, λ is a *self-conjugate partition*.
- The number of parts of the conjugate of a partition λ equals the largest part of λ.



Partition numbers V

- Let λ be the partition $n = n_1 + n_2 + \cdots + n_k$. More formally, the conjugate partition λ^* of λ is the partition $n = n_1^* + n_2^* + \cdots + n_l^*$ of n $(l = n_1)$ where n_i^* is the number of parts of λ which are at least equal to i,

$$n_i^* = |\{j : j \geq i\}| \qquad (i = 1, 2, \ldots, l).$$

Theorem

$$\sum_{n=0}^{\infty} p_n x^n = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$



Partition numbers VI

Proof

The expression on the right equals the product

$$(1 + x + \cdots + x^{1a_1} + \cdots)(1 + x^2 + \cdots + x^{2a_2} + \cdots)$$
$$(1 + x^3 + \cdots + x^{3a_3} + \cdots) \cdots .$$

A term x^n arises in this product by choosing a term x^{1a_1} from the first factor, x^{2a_2} from the second, x^{3a_3} from the third, and so on, with $1a_1 + 2a_2 + 3a_3 + \cdots = n$. Thus each partition of n contributes 1 to the coefficient of x^n, and the coefficient of x^n equals the number p_n of partition of n.

- Let $\mathcal{P}_n$ denote the set of all partitions of the positive integer n.



Partition numbers VII

- Let

$$\lambda : n = n_1 + n_2 + \cdots + n_k \qquad (n_1 \geq n_2 \geq \cdots \geq n_k)$$

and

$$\mu : n = m_1 + m_2 + \cdots + m_k \qquad (m_1 \geq m_2 \geq \cdots \geq m_k)$$

be two partitions of n. Then we say that λ is *majorized by* μ (or that μ *majorizes* λ) and write

$$\lambda \leq \mu.$$

provided the partial sums for λ are at most equal to the corresponding partial sums for μ

$$n_1 + n_2 + \cdots + n_i \leq m_1 + m_2 + \cdots + m_i \qquad (i = 1, 2, \ldots, k).$$



Partition numbers VIII

It is straightforward to check that the relation of *majorization* is reflexive, antisymmetric, and transitive, and hence is a partial order on $\mathcal{P}_n$.

Example

Consider the three partitions of 9

$$\lambda : 9 = 5 + 1 + 1 + 1 + 1; \quad \mu : 9 = 4 + 2 + 2 + 1; \quad \nu : 9 = 4 + 4 + 1.$$

For the purpose of comparing all three of these partitions we add trailing 0's to μ and ν, and think of μ as $9 = 4 + 2 + 2 + 1 + 0$ and ν as $9 = 4 + 4 + 1 + 0 + 0$. We have $\mu \leq \nu$ as

$$\begin{aligned} 4 &\leq 4; \\ 4 + 2 &\leq 4 + 4; \\ 4 + 2 + 2 &\leq 4 + 4 + 1; \\ 4 + 2 + 2 + 1 &\leq 4 + 4 + 1 + 0. \end{aligned}$$



Partition numbers IX

On the other hand, λ and μ are incomparable as $4 < 5$ but $4+2+2 > 5+1+1$. Similarly, λ and ν are incomparable.

- The lexicographic order can be also used on partitions to produce a total order of $\mathcal{P}_n$ that turns out to be a linear extension of the partial order of majorization.
- Let

$$\lambda : n = n_1 + n_2 + \cdots + n_k \qquad (n_1 \geq n_2 \geq \cdots \geq n_k)$$

and

$$\mu : n = m_1 + m_2 + \cdots + m_k \qquad (m_1 \geq m_2 \geq \cdots \geq m_k)$$

be two partitions of n. Then we say that there is an integer i such that $n_j = m_j$ for $j < i$ and $n_i < m_i$. Then we say that λ precedes μ in the *lexicographic order*, provided that there is an integer i such that $n_j = m_j$ for $j < i$ and $n_i < m_i$. It is simple to verify that lexicographic order is a partial order on $\mathcal{P}_n$.



Partition numbers X

Theorem

Lexicographic order is a linear extension of the partial order of majorization on the set $\mathcal{P}_n$ of partitions of a positive integer n.

- The fact that lexicographic order is a total (each two partitions of n are comparable) follows almost immediately from its definition.
- Let λ and μ be different partitions of n with λ majorized by μ Choose the first integer i such that $n_j = m_j$ for $j < i$ but $n_j \neq m_j$. Since

$$n_1 + n_2 + \cdots + n_{i-1} + n_i \leq m_1 + m_2 + \cdots + m_{i-1} + m_i,$$

we conclude that $n_i < m_i$, and hence λ precedes μ in the lexicographic order. □



A geometric problem I

- We shall obtain a combinatorial geometric interpretation of the sum

$$h_n^{(k)} = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{k} \quad (0 \le k \le n) \tag{8.21}$$

of the first $k+1$ binomial coefficients with upper argument equal to n–that is, the sum of the first $k+1$ numbers in row n of Pascal's triangle. For each fixed k, we obtain a sequence

$$h_0^{(k)}, h_1^{(k)}, h_2^{(k)}, \ldots, h_n^{(k)}, \ldots. \tag{8.22}$$

We use Pascal's formula to determine the differences of (8.22):

$$\begin{aligned}\Delta h_n^{(k)} =& h_{n+1}^{(k)} - h_n^{(k)} \\ =& \binom{n+1}{0} + \binom{n+1}{1} + \cdots + \binom{n+1}{k} \\ & - \binom{n}{0} - \binom{n}{1} - \cdots - \binom{n}{k}\end{aligned}$$



A geometric problem II

$$= \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{k-1}.$$

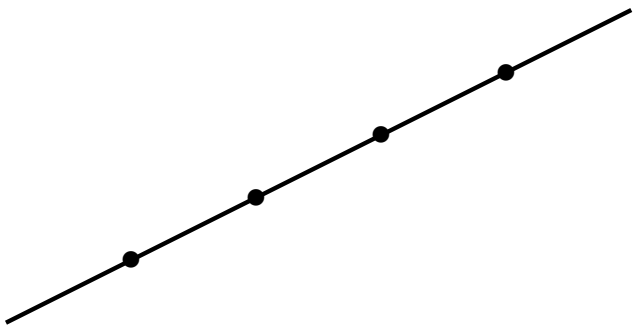
Hence

$$\Delta h_n^{(k)} = h_n^{(k-1)}. \tag{8.23}$$

- The number $h_n^{(k)}$ counts the number of combinations with at most k elements of a set with n elements. $h_n^{(k)}$ also has an interpretation as a counting function for a geometrical problem

  $h_n^{(k)}$ counts the number of regions into which k-dimensional space is divided by n $(k-1)$-dimensional hyperplanes in general position.

- $k = 0$, $h_n^{(0)} = 1$.
- $k = 1$, $h_n^{(1)} = n + 1$.



n points divide the line into $n+1$ regions ($n = 4$).

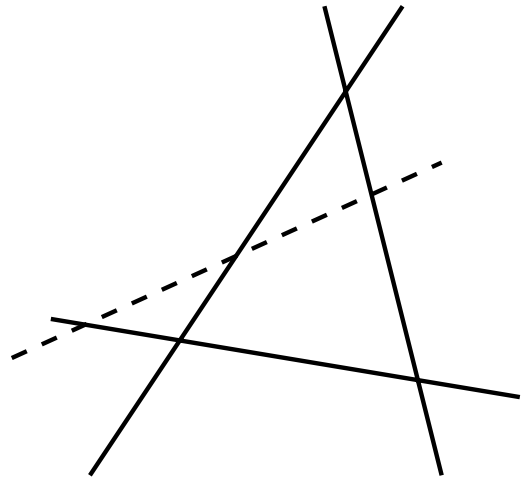
A geometric problem III

- $k = 2$, $h_n^{(2)} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} = \frac{n^2+n+2}{2}$ is the number of regions formed by n lines in general position in a plane.

Lines	0	1	2	3	4	5
Regions	1	2	4	7	11	16

- $k = 3$, $h_n^{(3)} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3}$ is the number of regions into which space is divided by n planes in general position in 3-space.
- The number of regions into which k-dimensional space is divided by n $k-1$-dimensional hyperplanes in general positions equals

$$h_0^{(k)}, h_1^{(k)}, h_2^{(k)}, \ldots, h_n^{(k)}, \ldots. \tag{8.24}$$



The new line (dashed) intersect the first n lines in n different points ($n = 3$). These n points divide the new line into $h_n^{(1)} = n + 1$ parts, each of them dividing a region formed by the first n lines into two.

Lattice paths and Schröder numbers I

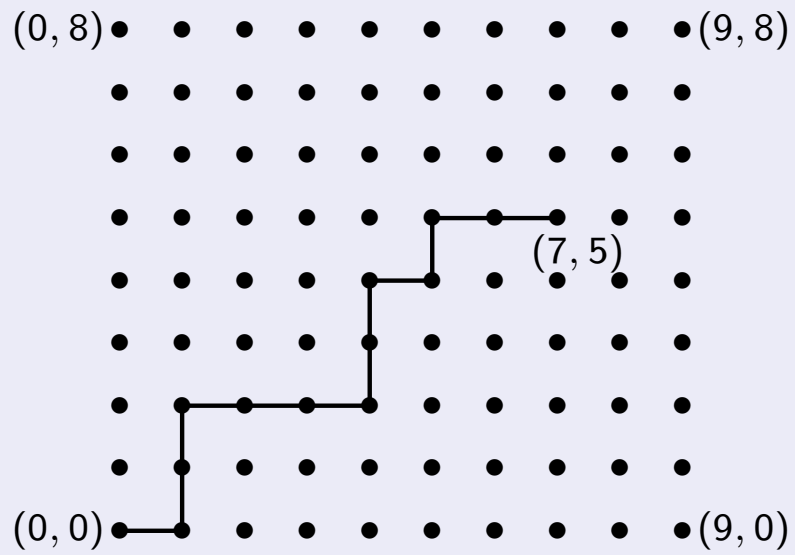
- Consider the *integral lattice* of points in the coordinate plane with integer coordinates. Given two such points (p, q) and (r, s), with $p \geq r$ and $q \geq s$, a *rectangular lattice path* from (r, s) to (p, q) is a path from (r, s) to (p, q) that is made up of *horizontal steps* $H = (1, 0)$ and *vertical steps* $V = (0, 1)$.

Example

In figure we show a rectangular lattice path from $(0, 0)$ to $(7, 5)$, consisting of 7 horizontal steps and 5 vertical steps. Given that the path starts at $(0, 0)$, it is uniquely determined by the sequence

$$H, V, V, H, H, H, V, V, H, V, H, H$$

of 7 H's and 5 V's.



Lattice paths and Schröder numbers II

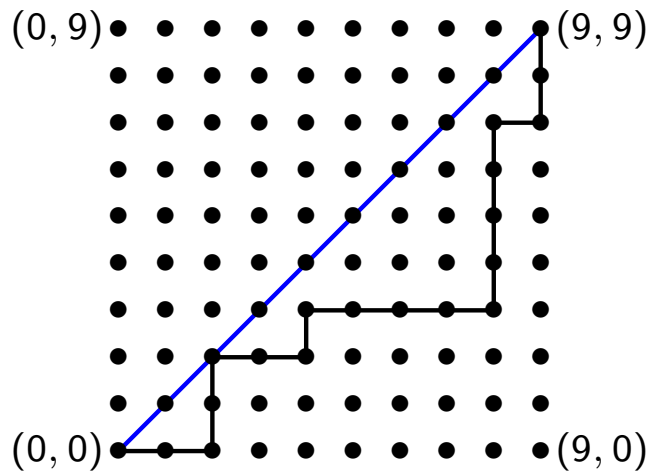
Theorem

The number of rectangular lattice paths from (r, s) to (p, q) equals the binomial coefficient

$$\binom{p-r+q-s}{p-r} = \binom{p-r+q-s}{q-s}.$$

- Without loss in generality, we'll consider rectangular lattice paths from $(0, 0)$ to (p, q), if $p \geq 0$ and $q \geq 0$, the number of rectangular lattice paths from $(0, 0)$ to (p, q) equals

$$\binom{p+q}{p} = \binom{p+q}{q}.$$



A subdiagonal path from $(0, 0)$ to $(9, 9)$.

Lattice paths and Schröder numbers III

- Consider rectangular lattice paths from $(0,0)$ to (p,q) that are restricted to lie on or below the line $y = x$ in the coordinate plane. We call such paths *subdiagonal rectangular lattice paths*.

Theorem

Let n be a nonnegative integer. Then the number of subdiagonal rectangular lattice paths from $(0,0)$ to (n,n) equals the nth Catalan number

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$



Lattice paths and Schröder numbers IV

- More generally, we can count the number of subdiagonal rectangular lattice paths from $(0,0)$ to (p,q) whenever $p \geq q$.

Theorem

Let p and q be positive integers with $p \geq q$. Then the number of subdiagonal rectangular lattice paths from $(0,0)$ to (p,q) equals

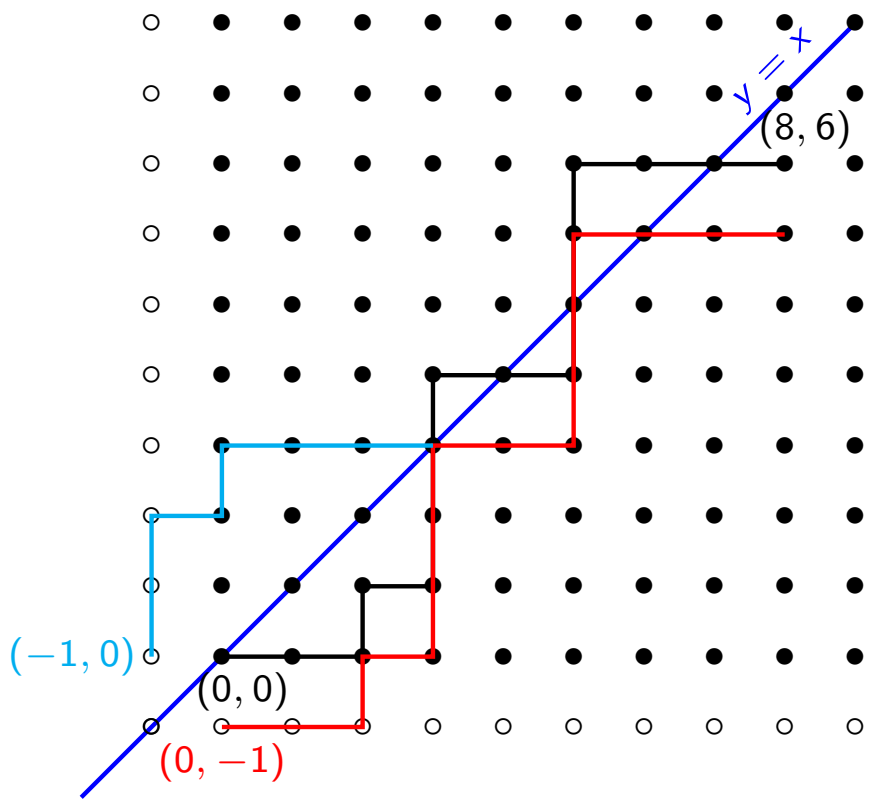
$$\frac{p-q+1}{p+1}\binom{p+q}{q}.$$

- We determine the number $I(p,q)$ of rectangular lattice paths γ from $(0,0)$ to (p,q) that cross the diagonal, and then subtract $I(p,q)$ from the total number $\binom{p+q}{q}$ of rectangular lattice paths from $(0,0)$ to (p,q).



Lattice paths and Schröder numbers V

- The number $l(p, q)$ is the same as the number of rectangular lattice paths γ' from $(0, -1)$ to $(p, q-1)$ that touch (possible cross) the diagonal line $y = x$. This follows by shifting paths down one unit, thereby shift a path γ into a path γ', and this establishes a one-to-one correspondence between the two kinds of paths.
- Consider a path γ' that touches the diagonal line $y = x$. Let γ'_1 be the subpath of γ' from $(0, -1)$ to the first diagonal point (d, d) touched by γ'. Let γ'_2 be the subpath of γ' from (d, d) to $(p, q-1)$.



Lattice paths and Schröder numbers VI

- We reflect γ_1' about the line $y = x$ and obtain a path γ_1^* from $(-1, 0)$ to (d, d). Following γ_1^* with γ_2, we get a path γ^* from $(-1, 0)$ to $(p, q-1)$.
- Now every rectangular lattice path θ from $(-1, 0)$ to $(p, q-1)$ must cross the diagonal line $y = x$. The correspondence γ' to γ^* is a one-to-one correspondence, and hence that $I(p, q)$ equals the number of rectangular lattice paths from $(-1, 0)$ to $(p, q-1)$.
- We have

$$I(p,q) = \binom{p+1+q-1}{q-1} = \binom{p+q}{q-1}.$$

Therefore, the number of subdiagonal rectangular lattice paths from $(0, 0)$ to (p, q) equals

$$\begin{aligned}\binom{p+q}{q} - I(p,q) &= \binom{p+q}{q} - \binom{p+q}{q-1} \\ &= \frac{(p+q)!}{p!q!} - \frac{(p+q)!}{(p+1)!(q-1)!}\end{aligned}$$



Lattice paths and Schröder numbers VII

$$= \frac{p-q+1}{p+1}\binom{p+q}{q}. \quad \square$$

- Now consider lattice paths for which, in addition to horizontal steps $H = (1,0)$ and vertical steps $V = (0,1)$, we allow *diagonal steps* $D = (1,1)$. Let p and q be nonnegative integers, and let $K(p,q)$ be the number of lattice paths from $(0,0)$ to (p,q) made up of horizontal, vertical, and diagonal steps, and let $K(p,q:rD)$ be the number of such paths that use exactly r diagonal steps D. We have

$$K(p,q:0D) = \binom{p+q}{p}.$$

We also have $K(p,q,rD) = 0$ if $r > \min\{p,q\}$.



Lattice paths and Schröder numbers VIII

Theorem

Let $r \leq \min\{p, q\}$. Then

$$K(p, q : rD) = \binom{p+q-r}{p-r\ q-r\ r} = \frac{(p+q-r)!}{(p-r)!(q-r)!r!},$$

and

$$K(p, q) = \sum_{r=0}^{\min(p,q)} \frac{(p+q-r)!}{(p-r)!(q-r)!r!}.$$



Lattice paths and Schröder numbers IX

- A lattice path from $(0,0)$ to (p,q) that uses r diagonal steps D must use $p-r$ horizontal steps H and $q-r$ vertical steps V, and is uniquely determined by its sequence of $p-r$ H's, $q-r$ V's, and r D's. Thus, the number of such paths is the number of permutations of the multiset

$$\{(p-r)\cdot H, (q-r)\cdot V, r\cdot D\},$$

which equals to

$$\frac{(p+q-r)!}{(p-r)!(q-r)!r!}.$$

If we do not specify the number r of diagonal steps, then by summing $K(p,q:rD)$ from $r=0$ to $r=\min(p,q)$, we obtain $K(p,q)$ as given in the theorem. □



Lattice paths and Schröder numbers X

- Now let $p \geq q$ and let $R(p, q)$ be the number of subdiagonal lattice paths from $(0, 0)$ to (p, q). Also, let $R(p, q : rD)$ be the number of subdiagonal lattice paths from $(0, 0)$ to (p, q) that use exactly r diagonal steps D. We have

$$R(p, q) = \sum_{r=0}^{q} R(p, q : rD).$$



Lattice paths and Schröder numbers XI

Theorem

Let p and q be positive integers with $p \geq q$, and let r be a nonnegative integer with $r \leq q$. Then

$$R(p, q : rD) = \frac{p-q+1}{p-r+1} \frac{(p+q-r)!}{r!(p-r)!(q-r)!}$$
$$= \frac{p-q+1}{p-r+1} \binom{p+q-r}{r\ p-r\ q-r},$$

and

$$R(p, q) = \sum_{r=0}^{q} \frac{p-q+1}{p-r+1} \frac{(p+q-r)!}{r!(p-r)!(q-r)!}.$$



Lattice paths and Schröder numbers XII

- We now suppose that $p = q = n$. The subdiagonal lattice paths from $(0, 0)$ to (n, n) are called *Schröder paths*. The *large Schröder number* R_n is the number of Schröder paths from $(0, 0)$ to (n, n). Thus,

$$R_n = R(n, n) = \sum_{r=0}^{n} \frac{1}{n - r + 1} \frac{(2n - r)!}{r!((n - r)!)^2}. \tag{8.25}$$

  The sequence $R_0, R_1, R_2, \ldots, R_n, \ldots$ of large Schröder numbers begins as

  $$1, 1, 2, 6, 22, 90, 394, 1, 806, \ldots.$$

- We now turn to the small Schröder numbers, which are defined in terms of constructs called bracketings. In a bracketing, a set of parentheses can enclose any number of symbols.



Lattice paths and Schröder numbers XIII

- For clarity, we shall remove parentheses around single elements as well as the last set of parentheses corresponding to the final bracketing of the remaining symbols.

Example

1. If $n = 1$, then there is only one bracketing, namely, a_1.
2. For $n = 2$, there is also one bracketing, namely, $a_1 a_2$.
3. For $n = 3$, we have 3 bracketings, $a_1 a_2 a_3$, $(a_1 a_2)a_3$, and $a_1(a_2 a_3)$.
4. For $n = 4$, we have 11 bracketings,

$$a_1 a_2 a_3 a_4, (a_1 a_2)a_3 a_4, (a_1 a_2 a_3)a_4, a_1(a_2 a_3)a_4, a_1(a_2 a_3 a_4), a_1 a_2(a_3 a_4)$$

and

$$((a_1 a_2)a_3)a_4, (a_1(a_2 a_3))a_4, a_1((a_2 a_3)a_4), a_1(a_2(a_3 a_4)), (a_1 a_2)(a_3 a_4).$$



Lattice paths and Schröder numbers XIV

- We now give the formal recursive definition of a bracketing of a sequence $a_1, a_2, \ldots, a_n$. Each symbol a_i is itself a bracketing; and any consecutive sequence of two or more bracketings enclosed by a set of parentheses is a bracketing. Using this definition, we can construct all bracketings of the sequence $a_1, a_2, \ldots, a_n$ by carrying out the following recursive algorithm in all possible ways.

Algorithm to construct bracketings

Start with a sequence $a_1, a_2, \ldots, a_n$.

1. Let γ equal $a_1 a_2 \ldots a_n$.
2. While γ has at least 3 symbols, do the following:
   1. Put a set of parentheses around any number $k \geq 2$ of consecutive symbols, say, $a_i \cdots a_{i+k-1}$, to form a new symbol $(a_i \cdots a_{i+k-1})$.
   2. Replace γ with the expression in which $(a_i a_{i+1} \cdots a_{i+k-1})$ is now one symbol.
3. Output the current expression.



Lattice paths and Schröder numbers XV

- A multiplication scheme for $a_1, a_2, \ldots, a_n$ is a *binary bracketing*–that is, a bracketing in which each set of parentheses encloses two symbols.

Example

A bracketing example of $n = 9$

$$\begin{aligned} a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 a_9 &\to a_1 a_2 a_3 (a_4 a_5 a_6) a_7 a_8 a_9 \\ &\to (a_1 a_2) a_3 (a_4 a_5 a_6) a_7 a_8 a_9 \\ &\to (a_1 a_2) a_3 ((a_4 a_5 a_6) a_7 a_8) a_9 \\ &\to (a_1 a_2)(a_3 ((a_4 a_5 a_6) a_7 a_8) a_9). \end{aligned}$$



Lattice paths and Schröder numbers XVI

- For $n \geq 1$, the *small Schröder number* s_n is defined to be the number of bracketings of a sequence $a_1, a_2, \ldots, a_n$ of n symbols. In fact, the sequence $\{s_n : n = 1, 2, 3, \ldots\}$ begins as

$$1, 1, 3, 11, 45, 197, 903, \ldots.$$

Theorem

The generating function for the sequence $\{s_n : n \geq 1\}$ of small Schröder numbers is

$$\sum_{n=1}^{\infty} s_n x^n = \frac{1}{4}(1 + x - \sqrt{x^2 - 6x + 1}).$$



Lattice paths and Schröder numbers XVII

- Let $g(x) = \sum_{n=1}^{\infty} s_n x^n$ be the generating function of the small Schröder numbers. The recursive definition of the bracketing implies that

$$\begin{aligned} g(x) &= x + g^2(x) + g^3(x) + g^4(x) + \cdots \\ &= x + g^2(x)(1 + g(x) + g^2(x) + \cdots) \\ &= x + \frac{g^2(x)}{1 - g(x)}. \end{aligned}$$

This gives

$$2g^2(x) - (1 + x)g(x) + x = 0.$$

The two solutions of this quadratic equation are

$$\frac{1 + x + \sqrt{x^2 - 6x + 1}}{4}, \qquad \frac{1 + x - \sqrt{x^2 - 6x + 1}}{4}..$$



Lattice paths and Schröder numbers XVIII

Since $g(0) = 0$, we have

$$g(x) = \frac{1 + x - \sqrt{x^2 - 6x + 1}}{4}. \quad \square$$

- A recurrence relation for the small Schröder numbers can be obtained as

$$(n+2)s_{n+2} - 3(2n+1)s_{n+1} + (n-1)s_n = 0, \quad (n \geq 1), \tag{8.26}$$

or

$$ns_n - 3(2n-3)s_{n-1} + (n-1)s_{n-1} = 0, \quad (n \geq 3),$$

which is a homogeneous linear recurrence relation of order 2 with nonconstant coefficients.



Lattice paths and Schröder numbers XIX

Theorem

The generating function for the sequence $\{R_n : n \geq 0\}$ of large Schröder numbers is

$$\sum_{n=0}^{\infty} R_n x^n = \frac{-(x-1) - \sqrt{x^2 - 6x + 1}}{2}.$$

A subdiagonal lattice path from $(0,0)$ to (n,n)

(i) is the empty path (if $n = 0$),
(ii) starts with a diagonal step D, or
(iii) starts with a horizontal step H.

- The number of paths of type (ii) equals the number of subdiagonal paths from $(1,1)$ to (n,n) and thus equals R_{n-1}.
- The paths of type (iii) begin with a horizontal step H and then follow a path γ from $(1,0)$ to (n,n) without going above the diagonal line joining $(1,1)$ and (n,n).



Lattice paths and Schröder numbers XX

- Since γ ends on the diagonal at the point (n, n), there is a first point (k, k) of γ on the diagonal, where $1 \leq k \leq n$.
- γ arrives at (k, k) by a vertical step V from the point $(k, k-1)$. The part of γ from $(1, 0)$ to $(k, k-1)$ is a lattice path γ_1 that does not go above the diagonal line joining $(1, 0)$ and $(k, k-1)$.
- The part of γ from (k, k) to (n, n) is a lattice path γ_2 that does not go above the diagonal line joining (k, k) and (n, n).
- There are R_{k-1} choices for γ_1 and R_{n-k} choices for γ_2, and hence the number of lattice paths of type (iii) equals $R_{k-1}R_{n-k}$.



Lattice paths and Schröder numbers XXI

- Summarizing, we get the recurrence relation

$$R_n = R_{n-1} + \sum_{k=1}^{n} R_{k-1} R_{n-k}, \quad (n \geq 1), \tag{8.27}$$

or equivalently,

$$R_n = R_{n-1} + \sum_{k=0}^{n-1} R_k R_{n-1-k}, \quad (n \geq 1), \tag{8.28}$$

where $R_0 = 1$. Thus,

$$x^n R_n = x(x^{n-1} R_{n-1}) + x \left(\sum_{k=0}^{n-1} x^k R_k x^{n-1-k} R_{n-1-k} \right), \quad (n \geq 1).$$



Lattice paths and Schröder numbers XXII

Since $R_0 = 1$, the preceding equation implies that the generating function $h(x)$ of the large Schröder numbers satisfies

$$h(x) = 1 + xh(x) + xh^2(x). \tag{8.29}$$

The two solutions of this quadratic equation for $h(x)$ are

$$\frac{-(x-1) \pm \sqrt{x^2 - 6x + 1}}{2x}.$$

The solution with plus sign does not give nonnegative integers so that

$$h(x) = \frac{1 - x - \sqrt{x^2 - 6x + 1}}{2x}. \quad \square$$



Lattice paths and Schröder numbers XXIII

Corollary

The large and small Schröder numbers are related by

$$R_n = 2s_{n+1}, \qquad (n \geq 1).$$

Theorem

Let n be a positive integer. Then the number of dissections of a convex polygonal region Π_{n+1} of $n+1$ sides equals the small Schröder number s_n.



Lattice paths and Schröder numbers XXIV

Algorithm to Construct Dissections of Π_{n+1}

Start with the convex polygonal region Π_{n+1}, with the sides labeled base, $a_1, a_2, \ldots, a_n$ in a clockwise fashion.

1. Let $\Gamma = \Pi_{n+1}$.
    1. While Γ has more than three sides, insert a diagonal of Γ, thereby partitioning Γ into two parts.
    2. Replace Γ with the part containing the base. (This part will have at least one fewer side.)
2. Output the full dissected polygonal region Π_{n+1}.



Lattice paths and Schröder numbers XXV

Example

$$
\begin{aligned}
& a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 a_9 \\
& \quad \to a_1 a_2 a_3 (a_4 a_5 a_6) a_7 a_8 a_9 \\
& \quad \to (a_1 a_2) a_3 (a_4 a_5 a_6) a_7 a_8 a_9 \\
& \quad \to (a_1 a_2) a_3 ((a_4 a_5 a_6) a_7 a_8) a_9 \\
& \quad \to (a_1 a_2)(a_3((a_4 a_5 a_6) a_7 a_8) a_9).
\end{aligned}
$$

